

Zhipeng Zhu

Data & AI Architect | Databricks Lakehouse | Enterprise AI
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SKILLS

DATA & AI PLATFORM

Databricks, Unity Catalog, Delta Lake
Spark / PySpark, Lakeflow
Vector Search, Metric Views

GENAI & AGENTS

Genie Agents, Genie One
Knowledge Assistants, Databricks Apps
RAG, Agent Evaluation

DATA ENGINEERING

Python, R, SQL, Shell, Git
AWS (S3), Azure Data Factory
Hive, Medallion / ETL-ELT

GOVERNANCE & IDENTITY

Unity Catalog governance
RBAC, ABAC, Okta SSO

DOMAIN

Enterprise Agent architecture
Governed memory vs experiential RAG
HITL eval / write-back loops

EDUCATION

MSC, STATS / DATA SCIENCE

UNIVERSITY OF BRITISH COLUMBIA
May 2022 | Vancouver, BC
Project & Co-op Double Track

BSC, PHYSICS / MATH & STATS

OHIO STATE UNIVERSITY
May 2020 | Columbus, OH
Magna Cum Laude

WRITING / OSS

DISTPLYR & DISTIONARY

R / PROBAVERSE
Distribution objects (`dst_*`);
dplyr-style shift, mix, maximize.

ZHUZP98.GITHUB.IO

First-principles Agent notes.
Five-module enterprise Agent loop.
Databricks implementation map.

EXPERIENCE

DATA & AI ARCHITECT | BITMART

Feb 2025 - July 2026 | Remote

Platform: In a three-month landing, personally built the production Databricks Lakehouse (200+ datasets, 60+ users, 9 BUs). Finance and Operations were live within two months. Genie, Metric Views, and Lark workflows cut the fastest ad-hoc requests by up to 20x, finance process hours by about 90%, and data-team attribution requests by about 50%. Development pace about 80% faster; maintenance and emergency events about 90% lower.

Delivery: Coordinated about 10 engineers across Information Security, Operations, and Data Platform for identity, source-database assessment, and legacy pipelines. Enabled the existing Okta SSO, with RBAC/ABAC and column-level masking. With the DBA team, delivered a one-year batch and realtime migration plan. Partial Superset, Doris, and Hue retirement is underway.

Cost: Reduced compute cost by about 40%. Weekly Genie usage reviews reduced Genie spend about 30-40%. Presented results to the president and senior management.

Integration: Published usage guides, engineering docs, and the data-security SOP on the company Lark Wiki, and synced Lark Wiki, Docs, and Sheets into Unity Catalog as department context. Replaced Superset dashboard and CSV push with Databricks Workflow Jobs.

DATA ENGINEER | BUDWEISER APAC

Jul 2024 - Dec 2024 | Shanghai

Architecture: Designed Databricks middle-platform solutions; rebuilt ODS - DW - ADS dependencies; optimized clusters and Azure Data Factory ETL.

Delivery: Partnered with supply chain and sales to model DWS metrics and multi-environment governance for risk-order inspection and business insight.

SELECTED PROJECTS

LARK × DATABRICKS INTEGRATION SOLUTION

Aug 2026 - Present | Open Source (MIT License)

Governed identity and bidirectional workflows: **Genie Bot** (OBO Genie One in Lark); **Bridge Bot** (org tenant token, Lark↔Databricks); **Workspace Add-on** (OBO Genie in Docs/Base); **Lark MCP** for Genie Code/One. MLflow evaluation and HITL write-back live here.

HIGH-FREQUENCY LIQUIDITY & EFFECTIVE SPREAD PLATFORM

July 2025 - July 2026 | BitMart

Analytics: Designed Quant Liquidity metrics (quoted/effective spread, LOB resilience via recovery / mean-reversion / depth refill, adequacy A→E→L) with As-of Joins of trades vs 15-min LOB; 35% lower average effective spread on core pairs.

Platform: Migrated legacy hive_metastore tables to Unity Catalog. Moved some classic jobs to Serverless, and moved the 15-minute order-book job from Serverless onto an existing classic cluster; liquidity cluster spend about 50% lower. BitMart spot books moved from REST to full WebSocket (184 to 1,407 pairs); other exchanges stayed on REST. A serial DAG of about 3 hours now finishes in under 30 min.